

Charles Babbage was a mathematician and computer pioneer. He designed huge calculating machines, the last of which—his Analytical Engine—is considered to be the forerunner of the modern computer. A lifelong inventor with many interests, he also promoted scientific societies and industrial mechanization.

MILESTONES

DIFFERENCE ENGINES

Begins designing his first calculating machine in 1819. Develops this into the second version in 1847.

GOLD MEDAL

Wins Astronomical Society's Gold Medal in 1824, for inventing a machine to calculate logarithms.

PRESTIGIOUS CHAIR

Made Lucasian Professor of Mathematics at Cambridge University in 1828, but does not give lectures.

MEETS LOVELACE

Becomes friends with Ada Lovelace in 1833. She later sees the potential of his Analytical Engine.

FINAL MACHINE

Designs most of his Analytical Engine by 1838; continues to refine it but never builds it.

Born in London to a wealthy banking family, Charles Babbage displayed a talent and passion for mathematics early on. As a child, he suffered from prolonged ill health, which disrupted his schooling. But he read widely and had learned much about modern mathematics by the time he entered Cambridge University in 1810. He was not impressed with the syllabus there, and this drove him to set up the Analytical Society in 1812, which aimed to promote innovative developments in mathematics.

Babbage was known for his provocative personality. Despite being a top mathematics student at his Cambridge college, he transferred to another one and graduated from there in 1814. He later became influential in scientific circles, and in 1816, he was elected a Fellow of the Royal Society of London. In 1820, he helped found the Royal Astronomical Society.

Manuals and machines for calculating

In the early 1820s, engineers, navigators, and mathematicians used logarithm tables for complex calculations. These tables were devised, copied, and typeset by hand—a process that was vulnerable to errors.

Babbage's Analytical Engine
was designed to be programmed using cards punched with holes. It is widely recognized as the precursor of the computer.

